

5 Assessment of the Impact of Folding Climate-Responsive Façade System in Energy Performance for the Future Adaptation and Evolution of Houses in Louisiana Climate Zone (Under Extreme Heat Conditions)

Year 4 (2023 Winter-Spring)
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Location: Baton Rouge, Louisiana
Duration: 4 Months
Softwares: Rhinoceros, Grasshopper (Ladybug), ClimateStudio, Lumion, Photoshop, Illustrator, Indesign

Climate change is a pressing issue globally causing extreme weather conditions, in this event, it imperative to develop buildings strategies that are resilient to such changes. As several studies have reported, extreme weather conditions are experienced in Louisiana through increased extreme heat events due to global warming (CDC, 2013). Additionally, urban areas are more impacted by heat due to urban heat island effects, disrupting thermal comfort and increasing cooling demand (Howe et al, 2019; Santamouris, 2020). This will lead to a further rise in carbon dioxide emissions, aggravating global warming, and resulting in a cycle (Freedman, 2022).

Who are the vulnerable groups to extreme heat?

- Minority ethnic groups in the United States are often less educated, more likely to be socio-economically disadvantaged and work in higher-risk occupations, and more likely to live in hazardous areas (Hansen, 2013). These areas often consist of neighborhoods that are deprived of parks, trees, and other green spaces, which adds to a higher ambient surface temperature. Moreover, the low-income population usually lives in low-thermal-quality houses and they are prone to exposure to high indoor and outdoor ambient temperatures while consuming more energy to satisfy thermal needs (Santamouris, 2020).
- Prolonged heat exposure is associated with heat stress by triggering acute heat-related illness or exacerbating pre-existing health conditions (Wondmagegn, 2019).

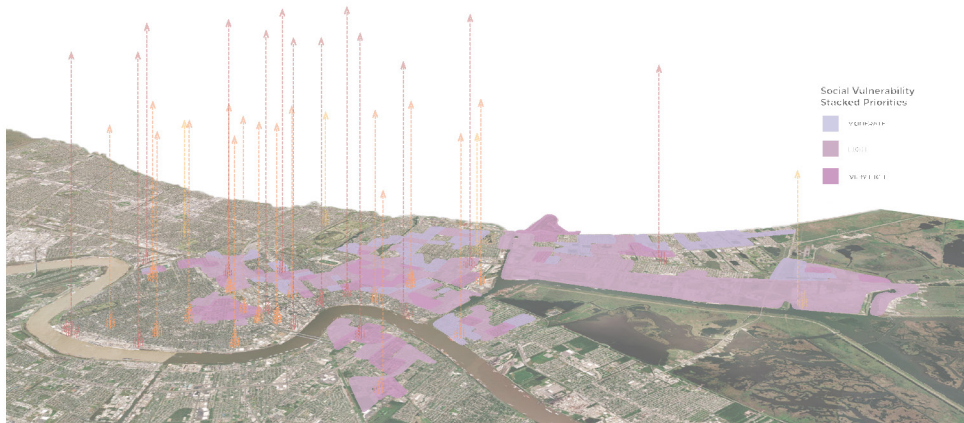
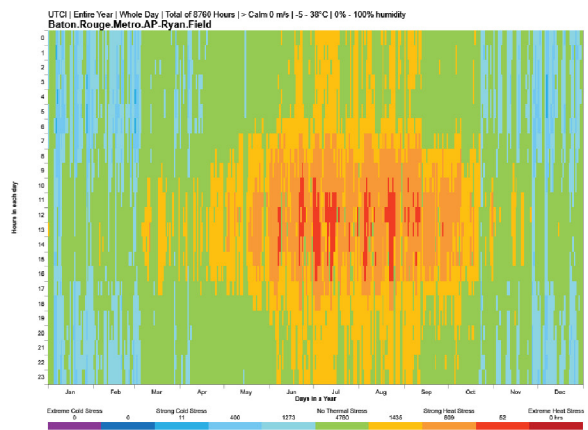


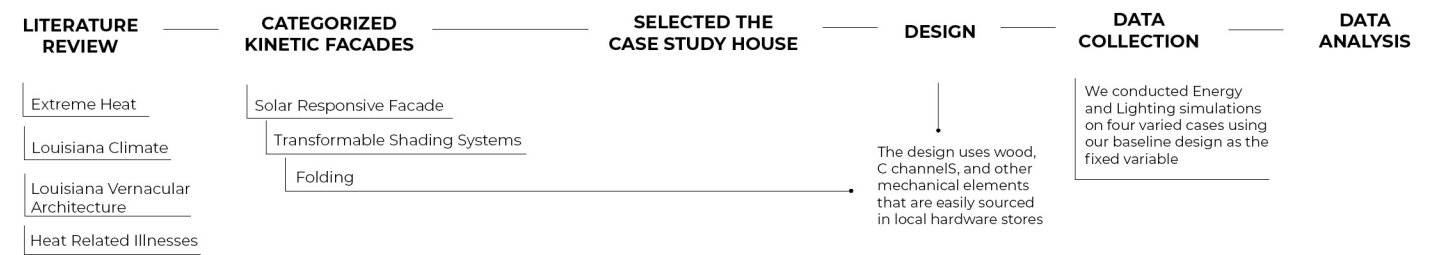
FIGURE 1
Urban Heat Island Effect & Socioeconomic Distribution : City of New Orleans
Stats source: NASA Applied Science Program, CDC Heat & Health Tracker
generated with ArcGIS, Adobe Suite



Thermal Stress Level: City of Baton Rouge
generated with Grasshopper

An adaptive and efficient building envelope enhances the energy performance of Louisiana buildings which will help reduce cooling demand in the long run. This study aims to investigate the energy performance efficiency of the folding climate-responsive facade system versus the existing static facade for shotgun houses in Louisiana. The strategy entails quantitative data for hourly shade states and the energy performance of the two facade states.

METHODOLOGY



With the goal of remediating the heat problem in residential buildings in Louisiana, the following measures are examined in the research:

- There are two major groups of kinetic facades: solar-responsive facades and wind-responsive facades (Abdelwahed Mekhamar & Halim Hussein, 2021). For the purpose of the research, the focus is on the solar kinetic facade with subgroups categorized by mechanical movements to steer the design specifically to target Louisiana heat issues (Figure 2).
- The shotgun house typology has been an efficient and economical option for Louisiana residents and has gained renewed interest following the trend of compact and sustainable living.
- Sustainable buildings that can adapt to changing climatic conditions explore an innovative approach to reduce energy consumption by integrating strategies, such as retrofitting existing buildings with more efficient envelopes as proposed in this research.

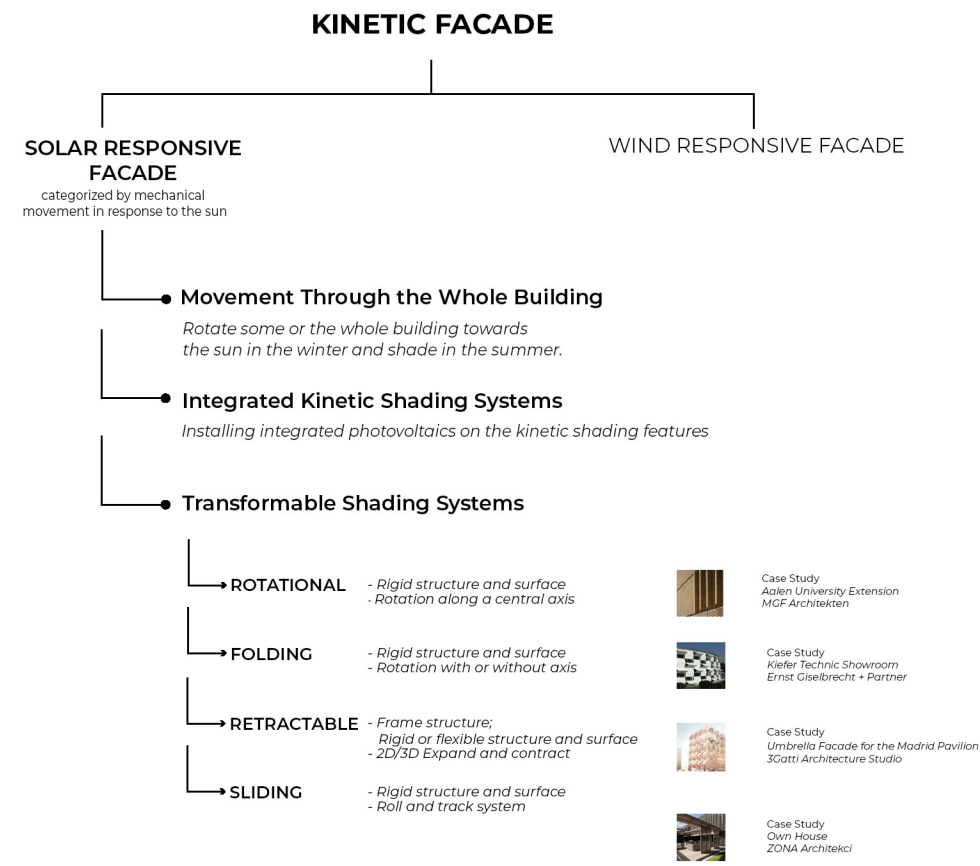
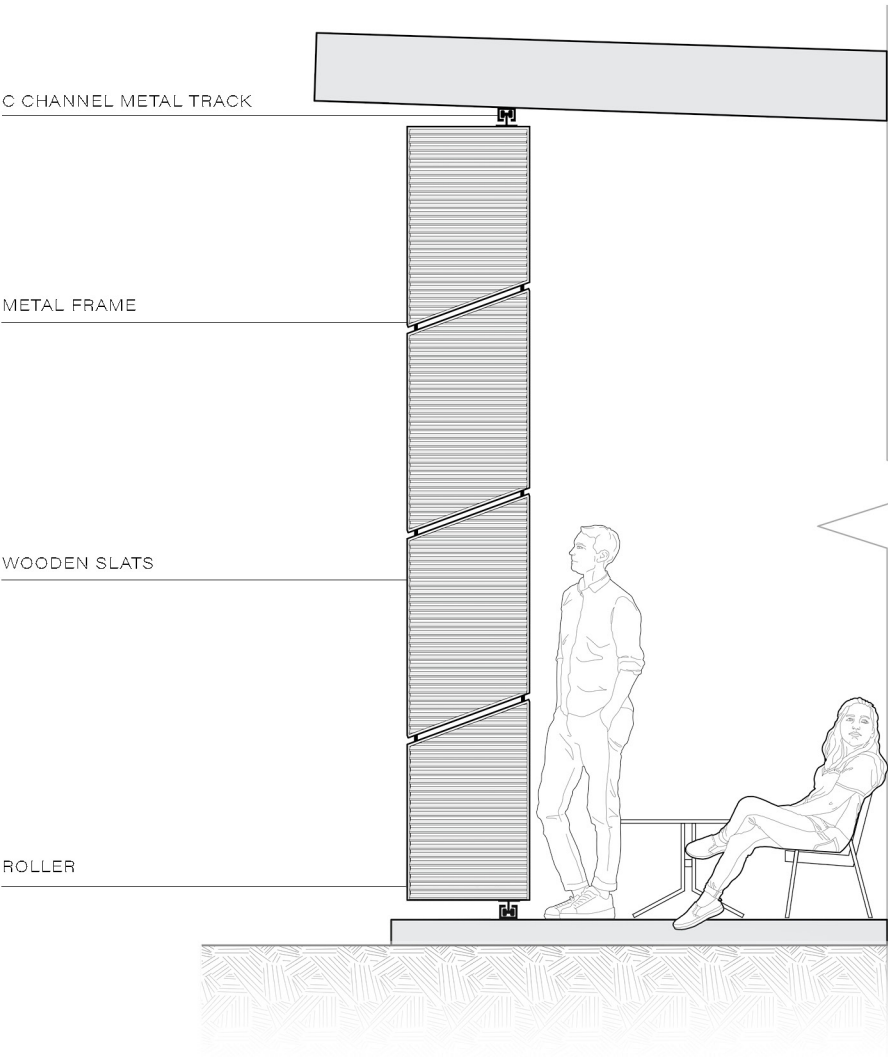
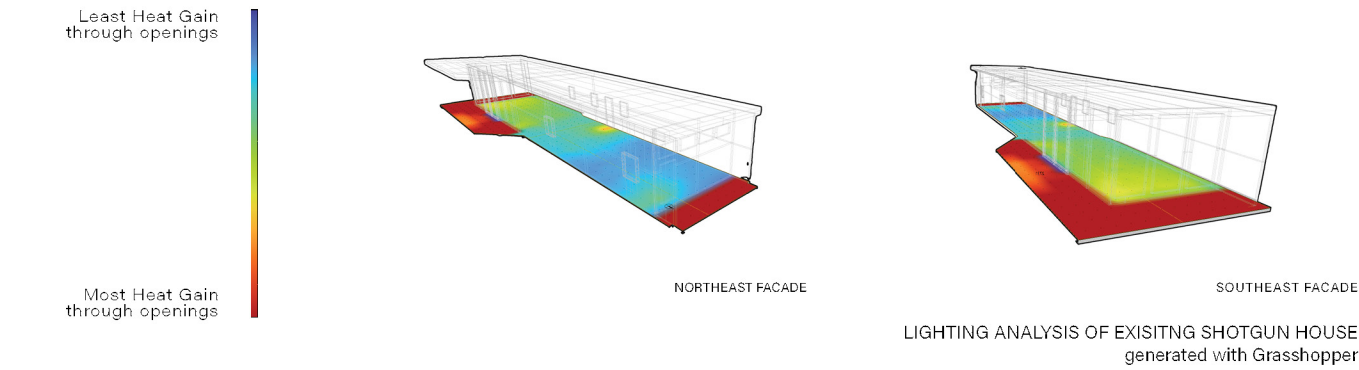
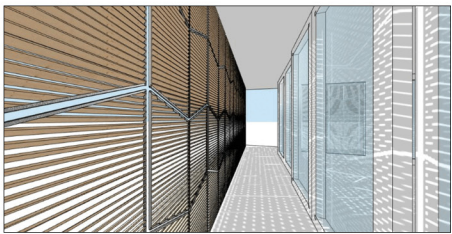
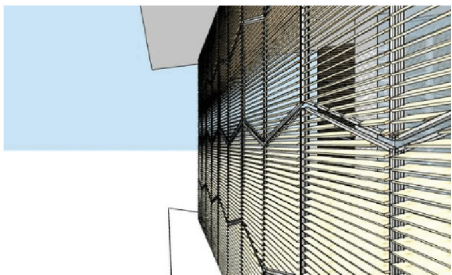


FIGURE 2
Kinetic Facade Categorization
generated with Illustrator

ANALYSIS & DESIGN

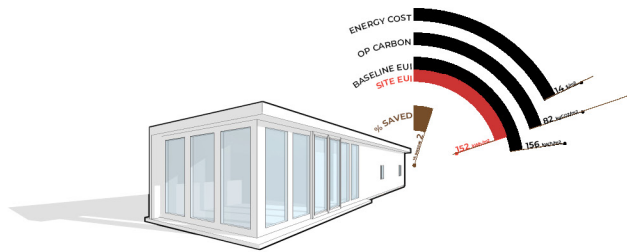


WALL SECTION ILLUSTRATING MECHANISM OF THE KINETIC SHADING SYSTEM DESIGN generated with Rhinoceros 3D, Illustrator



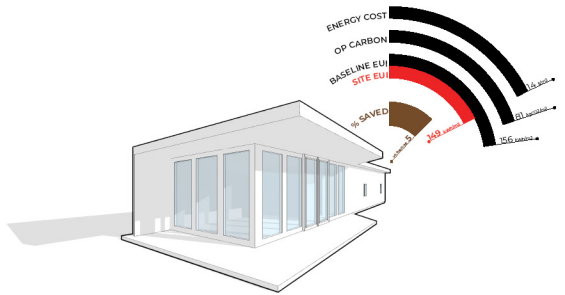
Case 1: Baseline (without overhang and shading system)

The overhang of the house was removed on the east facade to integrated more windows in order to maximize the chances for more daylight spaces inside. These modifications brought down the EUI to 152 kWh/m², which is lower than the baseline EUI, reducing energy costs and saving 2% of energy in comparison to the baseline EUI.



Case 2: With overhang and without shading system

The original overhang was integrated again into the baseline design which reduced the EUI further down to 149 kWh/m², saving 5% of energy in comparison to the baseline EUI. This demonstrates how efficient permanent overhangs on windows can be.



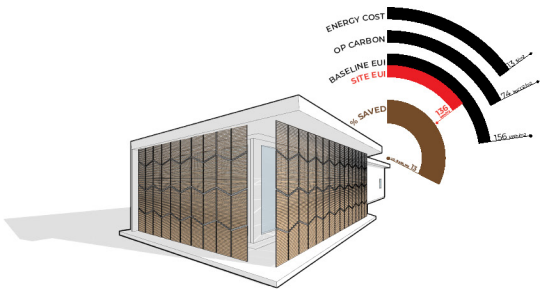
Case 3: With overhang and with shading system

In addition to the baseline design and overhang, kinetic shading system design was integrated on the south and east facades. With the unfolded kinetic facade, the EUI is 136 kWh/m², saving 13% of energy in comparison to the baseline EUI.

NOTES

EUI (Energy Use Intensity)

- EUI = Total Energy Consumed / Total Floor Area / Year
- Used to compare the energy efficiency of different buildings, track energy usage over time, and identify areas for improvement in a building's energy performance.



CONCLUSION

Through the analysis of three different shading scenarios, our research has shown that the kinetic shading system helped reduce the energy consumption of the house. Throughout the duration of this research, we learned that the east facade is the optimal location for shading systems, while the south facade is best suited for windows and overhangs. Additionally, the overhang had minimal impacts on the energy use of the building, but coupled with the shading system, the building's energy use decreased substantially.

For this research, extreme heat conditions was assumed for all simulation scenarios and focused on responding to the building's shading needs. However, a building's energy use is affected by other building components such as insulation, ventilation, lighting, etc. In the future, the research can be extended to the assessment of these other factors. Furthermore, this research could be expanded to a peak load study and/or a detailed material study to understand the impact and practicality of the proposed kinetic facade system.